

JUAN SEBASTIÁN PÉREZ

Senior AI Engineer · LLM Agents · RAG · Production AI Systems

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SUMMARY

Senior AI Engineer with 5+ years building software and 3+ years shipping LLM systems to production. I lead AI product engineering at AINEATECH (Houston, TX), where I designed and built 7 AI products for businesses in Colombia and Peru: WhatsApp AI sales agents for real estate, car dealerships and restaurants, a multi-tenant AI sales SaaS, and two AI CRMs (one with 120+ active users). I build reliable agents, not demos: strict tool schemas, deterministic guardrails, eval suites with LLM-as-judge, multi-tenant security and cost-aware model routing.

KEY ACHIEVEMENTS

- Designed a **reusable production agent architecture** (tool-use loop, strict-schema tools, deterministic guards, human handoff, Postgres RLS multi-tenancy, eval harness) and reused it across **4 verticals**. Launched a new vertical (car dealerships) on it in **2 days**.
- Architected a **production AI CRM with 120+ active users** across multiple companies in Colombia and Peru (RAG assistant, OCR invoices over WhatsApp, automated reporting).
- Eval-first engineering**: 1,500+ automated tests and 75+ eval scenarios (simulated customers, deterministic graders, LLM-as-judge). The real-estate agent's baseline showed **0 data leaks and 0 ungrounded figures**.

EXPERIENCE

Senior AI Engineer — Lead, AI Products

2023 – Present

AINEATECH (AINEA Technology LLC) · Houston, TX, USA · Remote

- Lead the end-to-end delivery of the company's AI products: architecture, LLM and prompt design, evals, security, CI/CD and production operations.
- Build agents as systems, not prompts: custom tool-use loops on the Claude Messages API with Zod strict-schema tools, structured outputs, versioned prompts and prompt caching. The agent cannot confirm an order or booking unless the tool call succeeded.
- Built deterministic output guards (data-leak canaries, price grounding against the catalog, policy checks) with one regeneration and a safe fallback, plus rule-based escalation to humans.
- Cost-aware LLM routing across Anthropic Claude (Opus/Sonnet) and open-weight models on Together AI (DeepSeek, Kimi, GLM), with fallbacks, per-turn cost tracking and benchmark-based model selection.
- Multi-tenant Postgres row-level security, AES-256-GCM encrypted credentials and "new client = configuration, not code" onboarding. Deep Meta integration: WhatsApp Cloud API, Coexistence, Embedded Signup, Conversions API, Marketing API.
- Production ops: Docker + Caddy on Hetzner, GitHub Actions CI/CD with automatic rollback, pg-boss queues, shadow-mode pilots with human review, ADRs and runbooks for every product.
- Also delivered 4+ production web and e-commerce platforms and n8n AI automations for clients (Pontebela, PrevenSalud, Grupo Alianza Vital).

SELECTED AI PRODUCTS

Sin Frontera — AI Sales Agent for B2B Real Estate (Colombia) **PILOT**

sinfrontera.aineatech.com

WhatsApp agent for an industrial real-estate firm (about 100 properties, 1,200–1,500 leads/month) that qualifies leads and books site visits. Hybrid retrieval combines SQL filters, Spanish full-text search and pgvector/Voyage embeddings, fused with RRF. It has 11 agent tools, plus a 12-tool internal assistant that lets the team query in natural language, make human-confirmed changes and get a daily summary. 43 eval scenarios with simulated prospects and an LLM judge, 540+ tests, 30 ADRs. Running as a supervised shadow-mode pilot.

TypeScript · Node 24 · Fastify · React · Supabase Postgres · pgvector · pg-boss · Claude · Together AI · Turborepo

Ainea — WhatsApp AI Sales Agent SaaS **DEPLOYED**

app.aineatech.com

Multi-tenant SaaS for LatAm SMBs that covers the full loop: Meta ad → WhatsApp chat → AI sales agent → store order → CAPI purchase event → repurchase. Self-serve onboarding with Embedded Signup and WhatsApp Coexistence. Shopify draft orders, Marketing API campaigns and Stripe Billing (3 plans). Each brand's agent prompt is generated from a vertical template, a brand brief and the catalog. Postgres turn queue with per-conversation locks, retries and a dead-letter queue. 760+ tests.

Ainea CRM — Sales Team CRM LIVE

app.aineatech.com

CRM for the sales team: leads with round-robin routing by product, clients, Google Meet scheduling, email campaigns with an automatic 3-step follow-up, metrics and commissions. "The app decides, n8n executes" through an HMAC-signed transactional outbox. Google SSO, 3 roles and a data-deletion flow for Colombia's Ley 1581.

Ainea Autos — AI Platform for Used-Car Dealerships (Colombia & Peru) DEPLOYED

autos.aineatech.com

Multi-tenant platform: a 24/7 WhatsApp buyer agent (inventory search, server-side financing calculator, trade-in capture, test-drive booking), a web CRM, and a 27-tool assistant for owners and salespeople that proposes actions for a person to confirm and sends proactive alerts. Deterministic lead-qualification engine; Google Sheets inventory sync.

Restaurant AI Agent — BACCA, 3 locations (Bucaramanga, Colombia) PILOT

Multi-tenant WhatsApp agent on Claude for delivery and pickup orders and table reservations. It has 13 strict tools and 4 deterministic guards (every price must match the menu or computed totals). Bookings respect seat capacity using Postgres advisory locks and slot holds. It handles voice notes (Deepgram) and extracts menus from photos and PDFs with Claude Opus. A new restaurant is onboarded with one command, as config rather than code. 240+ tests, 13 eval scenarios.

PrevenSalud AI CRM (Peru) LIVE · 120+ USERS

crm.prevensalud.pe

Production multi-tenant AI CRM: RAG assistant over live business data (no PII sent to the model), cost-first OCR invoice pipeline over WhatsApp (Tesseract → Vision → Claude) and AI-written PDF reports every night.

Python · FastAPI · PostgreSQL RLS · Redis · Celery · Next.js · Docker

Agentic AI Health Coach (mobile)

github.com/Juanllenato/plataforma-bienestar-ai

AI coach that runs an entire React Native app through chat. It orchestrates 13 tools: RAG on pgvector, vision-based meal logging and health projection. The backend is a modular FastAPI monolith with per-tenant feature flags.

Open source: LLM Eval Harness · Enterprise n8n AI Automations

github.com/Juanllenato

LLM evaluation harness with versioned datasets, LLM-as-judge and CI gating. n8n workflows for support triage, BANT lead scoring and invoice approval with fraud scoring and human-in-the-loop.

TECHNICAL SKILLS

AI / LLM: agentic systems and tool calling · RAG and hybrid search (pgvector, full-text, RRF) · structured outputs · context and prompt engineering · evals (LLM-as-judge, simulated users) · guardrails and prompt-injection defense · speech-to-text · OCR / document AI · Anthropic Claude · OpenAI · Together AI (DeepSeek, Kimi, GLM) · Voyage embeddings · LangChain/LangGraph concepts

Backend: TypeScript / Node.js (Fastify, Drizzle, Zod) · Python (FastAPI, async SQLAlchemy, Celery) · PostgreSQL / Supabase (RLS, pgvector) · Redis · pg-boss · REST and webhooks (idempotency, retries)

Integrations: WhatsApp Cloud API and Meta (Embedded Signup, CAPI, Marketing API) · Shopify · Stripe · Google Calendar / Sheets / Gmail · n8n

Frontend & DevOps: React · Next.js · React Native · Tailwind · Docker · Caddy / nginx · Linux VPS (Hetzner) · GitHub Actions CI/CD · Vitest · Playwright

EDUCATION

Systems Engineering — Unidades Tecnológicas de Santander (UTS) · In progress

Technologist in Software Systems Development — UTS · Completed

LANGUAGES

Spanish — Native

English — Intermediate (B1), improving; comfortable with written async communication (Slack, PRs, docs)